HDOC DAY-76

1. Create an enumeration for user roles (ADMIN, USER, GUEST) and display messages based on role.
    a. Input: GUEST
    b. Output: Welcome Guest
2. Print all enum names and integer values of 12 colors using a loop.
    a. Input: No Input
    b. Output: RED=0, BLUE=1, GREEN=2

1. Enumeration for User Roles:

Algorithm:

1. Start.
2. Define an enumeration Role containing ADMIN, USER, and GUEST.
3. Ask the user to enter a role:
    - 0 for ADMIN
    - 1 for USER
    - 2 for GUEST
4. Store the input in an integer variable.
5. Convert the entered value to the enum type.
6. Use a switch statement to check the role.
7. Display the corresponding welcome message.
8. Stop.

C Program:

```
#include <stdio.h>

enum Role {
  ADMIN,
  USER,
  GUEST
};

int main() {
  enum Role role;
  int choice;
```

```
    printf("Enter role (0-ADMIN, 1-USER, 2-GUEST): ");
    scanf("%d", &choice);

    role = (enum Role)choice;

    switch (role) {
        case ADMIN:
            printf("Welcome Admin\n");
            break;

        case USER:
            printf("Welcome User\n");
            break;

        case GUEST:
            printf("Welcome Guest\n");
            break;

        default:
            printf("Invalid Role\n");
    }

    return 0;
}
```

Test Case Table:

Test Case	Input	Expected Output
1	0 (ADMIN)	Welcome Admin
2	1 (USER)	Welcome User
3	2 (GUEST)	Welcome Guest
4	5	Invalid Role

Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case 1:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter role (0-ADMIN, 1-USER, 2-GUEST): 2
Welcome Guest
```

Test Case 2:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter role (0-ADMIN, 1-USER, 2-GUEST): 0
Welcome Admin
```

Test Case 3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter role (0-ADMIN, 1-USER, 2-GUEST): 1
Welcome User
```

Test Case 4:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter role (0-ADMIN, 1-USER, 2-GUEST): 5
Invalid Role
```

2. Print 12 Enum Color Names and Integer Values:

Algorithm:

1. Start.
2. Define an enumeration containing 12 colors.
3. Create a string array containing the corresponding color names.
4. Use a for loop from 0 to 11.
5. Print each color name and its enum integer value.
6. Stop.

C Program:

```
#include <stdio.h>

enum Color {
    RED,
    BLUE,
    GREEN,
    YELLOW,
    ORANGE,
    PURPLE,
    PINK,
    BROWN,
    BLACK,
    WHITE,
    GRAY,
    CYAN
};

int main() {
    char *colorNames[] = {
        "RED", "BLUE", "GREEN", "YELLOW",
        "ORANGE", "PURPLE", "PINK", "BROWN",
        "BLACK", "WHITE", "GRAY", "CYAN"
    };

    int i;

    for (i = RED; i <= CYAN; i++) {
        printf("%s = %d\n", colorNames[i], i);
```

```
    }


    return 0;
}
```

Test Case Table:

Test Case	Input	Expected Output
1	No Input	Print all 12 colors with values from 0 to 11

Compilation and Execution:

The following test case has been compiled and executed successfully.

Test Case 1:

